

# Rethinking journalism in the era of algorithms: Fairness, ethics, and the European regulatory framework

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As the adoption of AI in journalism is transitioning from basic automation toward autonomous decision-making, newsroom tasks are transformed, while traditional ethical and regulatory frameworks are challenged. News visibility is significantly mediated by algorithmic systems, necessitating a proactive approach to embed core journalistic values into automated editorial practices. This paper proposes a theoretical framework for journalism based on the dimensions of visibility, pluralism, and editorial autonomy as integral parts of algorithmic fairness. Through an interdisciplinary approach, the framework is structured on ethical impact assessment, technical evaluation, regulation compliance, and editorial oversight. It includes assessment of algorithmic impact on journalism before deployment, translation of the journalistic values into system constraints, and evaluation of algorithmic bias in news representation. It also ensures compliance with EU regulations like the AI Act, DSA, EMFA, and GDPR, while maintaining continuous human oversight to protect editorial independence and freedom of expression.

Keywords: Journalism, Media, Ethics, Fairness, Algorithms, Regulatory framework

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## 1 Introduction

The integration of artificial intelligence (AI) tools in journalism has deeply impacted a wide range of daily operations. Advancements in generative AI, especially in machine/deep learning (ML/DL) and natural language processing (NLP), have transformed professional tasks in the newsroom, not only triggering changes on the news production process but also challenging institutional approaches in the field of ethics and regulatory frameworks [1-3]. Furthermore, the adoption of agentic AI systems, which set goals, plan actions, make decisions, and execute tasks, signals a qualitative change in depth and not simply the more efficient completion of daily tasks [4]. The integration of AI in journalism is entering a new phase, where the focus is shifting from the automation of content production to decision-making and autonomy [5-7]. From search engines and social media feeds to automated recommendation systems and AI-assisted editorial tools, news production, moderation, and distribution are increasingly mediated by algorithmic systems [8-10]. Thus, decisions about news visibility are no longer made exclusively within editorial institutions, often leading to the exclusion of certain issues, regions, social groups, or viewpoints from the public discussion, thus highlighting the need to explore how news organisations embed journalistic values into practices [11, 12].

In this vein, the concept of algorithmic fairness has emerged in journalism not only as technical bias but as a challenge to freedom of expression, democratic discourse, and media plurality, as well. This paper explores how fairness and transparency should be implemented in journalism to ensure ethical, responsible, and trustworthy use of algorithms in the context of news ecosystems shaped by platform governance. It aims to contribute to the conceptual discussion of algorithmic fairness by proposing a theoretical framework that conceptualizes fundamental journalistic principles as four interconnected dimensions.

## 2 Artificial intelligence in journalism and emerging challenges

Media organizations adopt AI tools in their daily workflows to automate procedures during all phases of the news process. Such uses include –among others– document classification, language analysis, data cleaning, lead generation, and breaking news detection [13]. Newsrooms also implement AI technologies to discover, gather, and product news stories, to transform, manage, and distribute content, and to interact with their audience [14].

Furthermore, agentic large language models (LLMs) have emerged as valuable workflow accelerators [15], since they can contribute to the evaluation of articles or the summary of long reports, thus optimizing the editorial process [16]. In this direction, Maltezos et al. (2025) [15] developed Teleflash, a user-centered tool adapted to journalists' needs, especially in pressured situations, such as covering conflicts. It is a computational system designed to help journalists in the automated collection, filtering, and summary of information from dozens of channels on Telegram. Initially, the researchers mapped journalists' needs through interviews. Then, the specific reporting system was developed following a modular architecture, which consists of a data collection unit via the Telegram API, a filtering unit using patterns in multiple languages, a summarization unit that uses OpenAI's GPT models to generate reports, and a distribution unit that channels the reports into a Slack environment. Pilot testing with journalists showed that TeleFlash was able to efficiently extract relevant content from huge volumes of data, however the implementation also highlighted weaknesses, such as difficulties in verifying the reliability of sources and errors due to lack of context or "hallucinations" of the model.

In Sweden, the newspaper Svenska Dagbladet has begun experimenting with an AI-generated research team, following interviews with some of its existing subscribers, as well as some who are likely to join in the future. The basic idea is that editors will be able to consult these always-available digital twins on text editing or headline suggestions. At the same time, the Times of London used a synthetic research team to test the name of a new business podcast [4].

At the same time, though, scholarship increasingly focuses on the inherent challenges associated with algorithmic curation in journalism and, particularly, the opaque nature of internal algorithms that control a series of journalistic processes. Lack of transparency along with the complexity and adaptability of algorithmic functions makes it difficult to promote accountability and to evaluate them vis-à-vis ethical models [17]. News selection and visibility are significantly influenced by recommendations systems, which can amplify or suppress stories in user feeds and search results [9, 18, 19] building personalized information environments, while challenging fundamental journalistic concepts such as gatekeeping and public sphere [20, 21]. At the same time, the adoption of algorithms by media organizations to moderate users' content raises questions regarding legitimacy and proportionality as core functions of press freedom [22, 23].

### 3 Proposed theoretical framework

Against this background, algorithmic fairness in journalism cannot be considered a mere technical issue but also an ethical and democratic concern. Although fairness constitutes a concept that has been long debated in the journalistic field [e.g. 24, 25], the dominance of algorithmic tools underscores the need for a collective multidimensional approach. Specifically, visibility, pluralism, and editorial autonomy should be considered the main branches of algorithmic fairness in the field of journalism. At first, all involved actors (journalists, users, and media organizations) should have access to interpretable and explicable data about the influence of algorithmic systems on news visibility to enable accountability. Second, the design, training, and incorporation of such models in journalism should support exposure to diverse viewpoints and prevent systematic narrowing of public discourse. Last, human oversight should remain the core around which the journalistic function is structured, ensuring that editorial judgment is not overridden by platform optimization logic [26]. Together, these principles reposition algorithmic fairness as a structural requirement for democratic journalism in news platforms.

More specifically, visibility ensures the transparency and explainability of algorithmic decisions that determine the selection, ranking, and distribution of news content, enabling journalists and audiences to understand how editorial outcomes are influenced by automated systems [27]. Pluralism encompasses inclusion of diverse sources, viewpoints, and social groups, thereby supporting informed public debate and reducing the risk of algorithmically reinforced information echo chambers [28, 29]. Editorial autonomy emphasizes that AI systems should support rather than replace journalistic decision-making, preserving professional judgment, accountability, and editorial responsibility during the news production process [30].

Building on these three normative principles, the proposed framework operationalizes algorithmic fairness into four dimensions, acting as a multi-layered, interdisciplinary protocol. It integrates technical evaluation with social and legal requirements to address the democratic role of journalism in the AI-mediated media landscape. In particular, the proposed components are the ethical impact assessment, the technical evaluation, the regulatory alignment, and the editorial judgment.

In the *ethical impact assessment* dimension, a qualitative evaluation of how the algorithmic system could impact the news value chain takes place. Before the system is fully deployed in the workflow, core journalistic values, such as accuracy

and impartiality, have to be translated into algorithmic constraints [31]. Ethical frameworks for trustworthy and responsible integration of AI in journalism provide concrete guidelines for interpreting fundamental journalistic principles into operational requirements [32, 33]. For example, when emphasizing the role of journalism as a public watchdog, an AI-powered recommendation system could be assessed for whether it prioritizes engagement at the expense of pluralism.

The dimension of *technical evaluation* aims to address the “black-box” problem by employing bias mitigation techniques to ensure transparency [34]. Focus should be placed on exploring how algorithmic inputs and outputs influence the representation of diverse topics, groups, and news stories. To this direction, recent research proposes the incorporation of a training component on AI bias in a co-creation process with vulnerable and marginalized participant groups as a means for making AI bias addressing more accessible. Such a process, which was applied by socio-technical researchers, can empower journalists as end-users to be more aware of how AI bias can impact them and to recognize where there is a risk of discrimination occurring [35].

The orientation of *the regulatory alignment* dimension is to establish the implementation of the European legislative framework throughout the news production process [36]. Specifically, obligations posed by the European Artificial Intelligence Act (EU AI Act) to label, for example, AI-generated content and avoid practices that could cause negative effects on democratic processes could support beneficial outcomes for media organizations. The Digital Services Act (DSA) supports freedom and pluralism of the media through algorithmic transparency and content moderation provisions, posing rules to mitigate large-scale risks to society and democracy on online platforms, in full respect of users’ freedom of expression. Also, the European Media Freedom Act (EMFA) explicitly aims to safeguard editorial independence and media pluralism. Finally, the General Data Protection Regulation (GDPR) should map the auditing procedure in terms of privacy issues and prevention of data infringements.

The *editorial judgment* dimension underscores the requirement for continuous monitoring to protect journalistic autonomy against threats to freedom of expression that may be caused of automated decisions. Human moderation is a crucial necessity throughout the whole process. Although mentioned as a separate section of the proposed framework, this dimension functions as a horizontal governance requirement across the entire framework, ensuring that editorial judgment, responsibility, and accountability are maintained throughout all stages of the AI integration. Human-in-the-loop is also of vital importance for journalists themselves to ensure quality and professional standards when incorporating AI tools in their daily professional activity [37].

## 4 Discussion and Conclusions

The intersection of AI ethics, media studies, and law shapes the development of algorithmic fairness in journalism. Based on the fundamental professional values, media organizations need to extend the concept beyond technical definitions. News production, editing, and distribution is reframed as a fairness-sensitive algorithmic process under an interdisciplinary umbrella of technical requirements, ethical guidelines, and regulatory standards. Debates about effective platform governance should be inextricably linked with media pluralism and the public role of journalism in contemporary democracies. As credibility is crucial for journalism, discussion can focus on how AI could potentially enhance and strengthen the trust relationship between audiences and news media [38], which lies at the lowest level recorded in Reuters Institute Digital News Reports during the last decade [39].

As journalism becomes increasingly dependent on algorithmic infrastructures, questions of fairness extend beyond individual decision outcomes to the systemic organization of public visibility. Algorithmic systems not only distribute news but they actively shape the epistemic structure of public discourse. Reconceptualizing fairness in journalism therefore requires moving toward a broader normative framework that incorporates pluralism, autonomy, and transparency as foundational principles of algorithmic governance in news ecosystems.

However, the proposed framework has several limitations. First, it requires further operationalization into specific implementation guidelines, to support adoption by journalists and media organizations. Such guidelines could constitute a toolkit along with practical tools and measurable indicators. Furthermore, future research should explore how the framework can be empirically validated through real-world case studies and newsroom applications. Its effectiveness in identifying and mitigating algorithmic bias in practice could thus be tested across media organizations with different characteristics and AI-enabled journalistic workflows. Finally, while the framework establishes normative principles for algorithmic fairness, attention has to be paid to interpret these principles into standardized auditing methodologies, evaluation metrics, and best practices that can be systematically implemented and compared across diverse journalistic contexts.

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